INCH-POUND
MIL-DTL-3155D(SH)
10 September 2013
SUPERSEDING
MIL-V-3155C(SHIPS)
19 August 1964

DETAIL SPECIFICATION

VALVE, TEMPERATURE-REGULATING VENTILATION HEATER, STEAM CONTROL, NAVAL SHIPBOARD USE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers thermostatic-controlled steam valves for temperature regulation of shipboard ventilation heaters. These valves are of the self-operated type, requiring no separate power source such as electricity or compressed air.

1.2 Classification. Temperature-regulating valves with thermostat types are of the following models, as specified (see 6.2):

a. Model G or Model E with one of the following thermostats:
   (1) Weather (Type W)
   (2) Duct (Type L)
   (3) Room (Type R)

b. Model D with one of the following combinations of thermostats:
   (1) Type W and Type L
   (2) Type W and Type R

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4820

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:13Z
Check the source to verify that this is the current version before use.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-S-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-F-1183 - Fittings, Pipe, Cast Bronze, Silver-Brazing, General Specification for

(Copies of these documents are available online at http://quicksearch.dla.mil/ or https://assist.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available from the American National Standards Institute, 25 W. 43rd Street, 4th Floor, New York, NY 10036 or online at http://webstore.ansi.org/.)

ASTM INTERNATIONAL

ASTM A276 - Standard Specification for Stainless Steel Bars and Shapes

ASTM B61 - Standard Specification for Steam or Valve Bronze Castings

ASTM B584 - Standard Specification for Copper Alloy Sand Castings for General Applications

(Copies of these documents are available from ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The temperature-regulating valves furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.2).

3.2 Description.

3.2.1 Temperature-regulating valve. The temperature-regulating valve shall consist of the basic valve body model, valve seat(s) and valve poppet(s), and the thermostat type(s) necessary to accomplish the designated function.

3.2.2 Thermostat. A thermostat shall consist of steam actuated bellows, union nut, armored capillary tubing, and bulb.

3.3 Materials.

3.3.1 Temperature-regulating valve materials. Materials used in the temperature-regulating valve shall be high-grade, free from defects, and suitable for the purpose intended.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:13Z
Check the source to verify that this is the current version before use.

3.3.2 Joint materials. Silver brazing alloy or phosphorous copper brazing alloy shall be used for joints subjected to the temperature of live steam.

3.3.3 Dissimilar metals. Associated materials shall be compatible under intended service conditions.

3.3.4 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

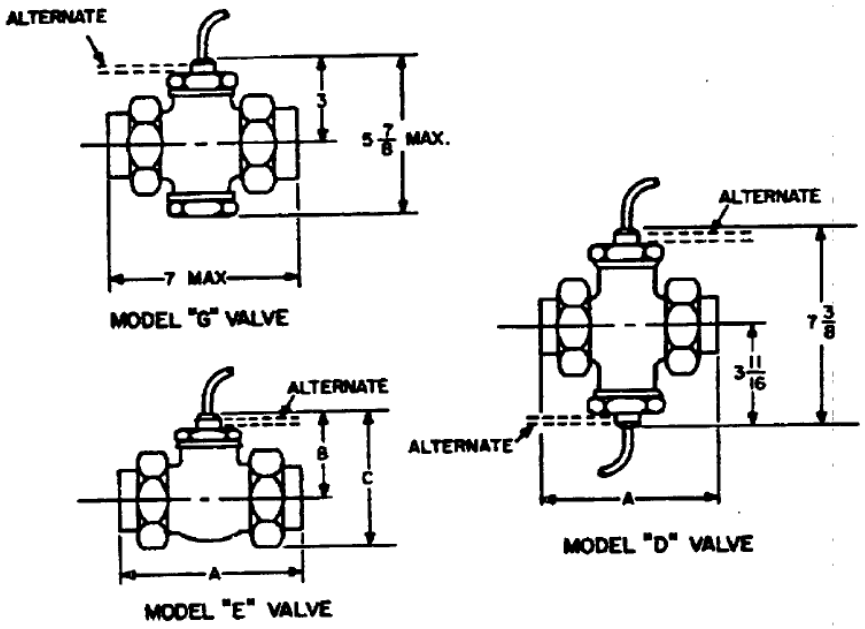
3.4 Design.

3.4.1 Removable parts. The temperature-regulating valve shall be designed so all interior parts of the valve can be removed for servicing without removing the valve body from the line, which includes replacement of the thermostat.

3.4.2 Shock. The temperature-regulating valve shall be designed such that it is capable of passing the high-impact (H.I.) shock test in accordance with MIL-S-901 for Grade A, Type A, Class I equipment.

3.4.3 Dimensions and tolerances. The physical dimensions and tolerances of the temperature-regulating valve shall conform to figure 1 for the models and sizes specified in table I and table II, as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:13Z
Check the source to verify that this is the current version before use.



Model and size number	Dimensions in inches (maximum)		
	A	B	C
E-½	4⅞	2	3
E-1	4⅞	2	3
E-2	6	3	4
E-3	6	3	4
E-4	6⅜	3	4
E-5	6⅜	3	4
E-6	6⅜	3	4
E-7	6 1/16	3	4
E-8	6 1/16	3	4
E-9	6½	3	4
E-10	6½	3	4
E-11	7	3	5
E-12	7	3	5

Model and size number	Dimensions in inches (maximum)
	A
D-21	6½
D-22	6½
D-23	6½
D-24	6⅛
D-25	6⅛
D-26	7
D-27	7
D-28	7½
D-29	7½

FIGURE 1. Valve dimensions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:13Z
Check the source to verify that this is the current version before use.

TABLE I. Models E and G valve sizes and capacities.

Model and size number	Nominal pipe size (I.P.S)	Capacity in pounds steam per hour	
		Minimum	Maximum
E-½	½	5	6.5
E-1	½	10	13
E-2	½	15	19
E-3	½	22	27.5
E-4	¾	30	37.5
E-5	¾	40	50
E-6	¾	60	75
E-7	1¼	90	112.5
E-8	1¼	120	144
E-9	1½	160	192
E-10	1½	200	240
E-11	1½	260	312
E-12	1½	325	390
G-32	1½	450	495
G-33	1½	600	660
G-34	1½	900	990

TABLE II. Model D valve sizes and capacities.

Model and size number	Nominal pipe size (I.P.S)	Capacity in pounds steam per hour			
		Type W –thermo		Type L or R –thermo	
		Minimum	Maximum	Minimum	Maximum
D-21	¾	10	15	30	37.5
D-22	¾	15	19	40	50
D-23	¾	22	27.5	60	75
D-24	1	30	37.5	90	112.5
D-25	1	40	50	120	144
D-26	1¼	60	75	160	192
D-27	1¼	90	112.5	200	240
D-28	1½	90	112.5	260	312
D-29	1½	120	144	325	390

3.4.4 Pressure capacity. Capacity of each model and size valve, based on a 35-pound per square inch gage (psig) inlet pressure and a 25-psig outlet pressure with the valve fully open, shall be in accordance with table I and table II, as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:13Z
Check the source to verify that this is the current version before use.

3.4.5 Valve operation. The valves, except Model D, shall be operative by any of the thermostats specified in 3.4.7. The Model D valve shall be operative by a combination of the Type W thermostat and the Type L or Type R thermostat, as specified (see 6.2).

3.4.6 Valve unit.

3.4.6.1 Packless type valve. Valves shall be of the packless type. The use of stuffing boxes or packing glands on valve stems shall not be permitted.

3.4.6.2 Valve bodies material. Valves bodies shall be made of bronze conforming to ASTM B61 or copper alloy UNS No. C92200 of ASTM B584. Valves shall have union ends in accordance with MIL-F-1183 with pre-inserted rings.

3.4.6.3 Valve seats and poppets materials. Valve seats and poppets shall be of corrosion-resistant steel equal to that of Type 303 of ASTM A276 and shall be removable and replaceable. The poppets for Types R and L thermostats shall have an orifice having throttling characteristics. The throttling characteristics shall be such that greater increments of valve poppet movement are required per pound of steam near the closed position than are required near the open position.

3.4.7 Thermostat. The steam flow of the valves shall be controlled by a bulb thermally responsive and connected to the steam operating bellows of the valve by a capillary tubing of the lengths specified in 3.4.9.1.

3.4.7.1 Interchangeability. The thermostat shall be connected to the basic valve body by a ring nut and shall be interchangeable as specified in table III.

TABLE III. Interchangeability of thermostats.

Model of valves	Sizes
E	½ through 1
E	2 through 7
E	8 through 12
D	21 through 24
D	25 through 29
G	32 through 34

3.4.7.2 Temperature shock. Each thermostat shall be subjected to an air temperature of not less than 180 °F for not less than 72 continuous hours without detrimental effect to any part of the element.

3.4.7.3 Type W thermostat. The thermostat shall be non-adjustable and shall be set at the factory to be fully open in a 35 °F ambient temperature.

3.4.7.4 Types L and R thermostat. The thermostats shall be provided with a dial adjustment graduated in degrees Fahrenheit. Means shall be provided for locking the dial at any set point within the operating range specified in table IV.

TABLE IV. Thermostat ranges.

Type	Range °F
L	40 to 100
R	60 to 80

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:13Z
Check the source to verify that this is the current version before use.

3.4.7.4.1 Throttling range. A 1 °F change in either direction of the thermostat set point of the ambient air temperature shall be sufficient to start the valve poppet through its operating stroke and the operating stroke shall be sufficient to ensure complete throttling or modulating operation. The throttling range between fully open and fully closed shall not exceed 6 °F.

3.4.7.4.2 Thermostat tolerance. The thermostats shall be capable of operating the valve controlled from fully open to fully closed with a change in air temperature at the thermostat bulb of not more than ±3 °F from the thermostat set point.

3.4.8 Bellows. Bellows shall be of the seamless type. The operating bellows of the steam valve shall be made of nickel-copper or of phosphor-bronze and shall be protected from direct steam impingement by a shield.

3.4.9 Capillary tubing. The capillary tubing shall be covered throughout the entire length with an armor to provide physical protection. Armor shall be brass, bronze, or corrosion-resistant steel so constructed that the flexibility of the tubing will not be excessively reduced.

3.4.9.1 Capillary tubing length. The length of the capillary tubing between the bulb and operating bellows shall be 10 feet for Type L and Type W thermostats and 25 feet for Type R thermostat.

3.4.10 Bulbs.

3.4.10.1 Bulb materials and lengths. Thermostat bulbs shall be constructed of brass or copper and the bulb lengths shall not exceed the dimensions shown on figure 2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:13Z
Check the source to verify that this is the current version before use.

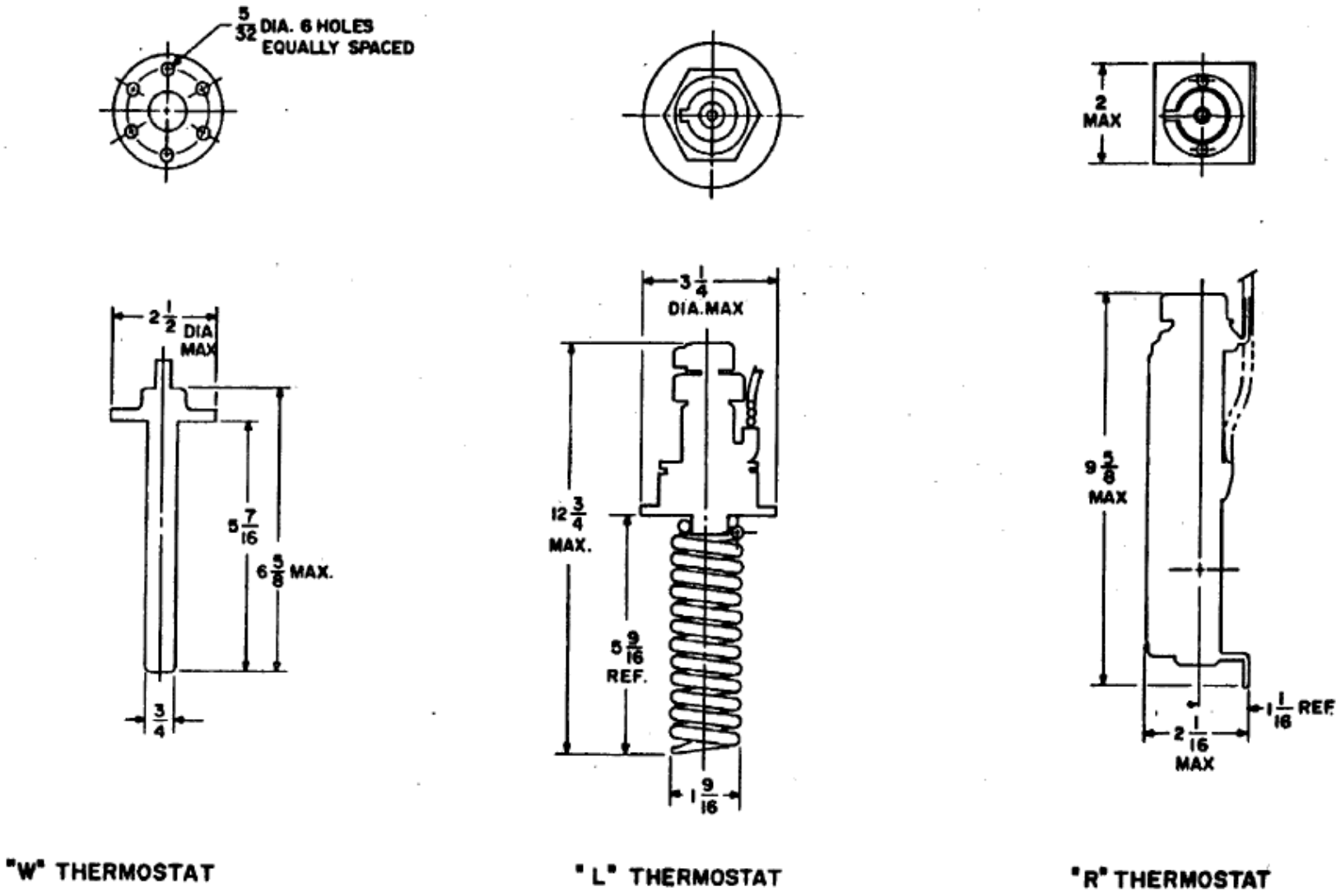


FIGURE 2. Thermostat dimensions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:13Z
Check the source to verify that this is the current version before use.

3.4.10.2 Bulb mounting. Bulbs shall have means for mounting similar to that shown on figure 2 and the mounting frame or flange shall be drilled for mounting fasteners.

3.5 Manually operable element. When specified (see 6.2), a manually operable control element capable of being inserted in the valve body instead of a thermostat shall be furnished. The element shall be capable of manually controlling the flow of steam from closed position to fully open, and shall have an index marking when practicable to show the relative position of the valve poppet with respect to its seat.

3.6 Bulkhead stuffing box. When specified (see 6.2), a bulkhead stuffing box for passing the capillary tubing through watertight bulkheads or decks shall be provided. The stuffing box shall be constructed of brass and shall be in accordance with figure 3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:13Z
Check the source to verify that this is the current version before use.

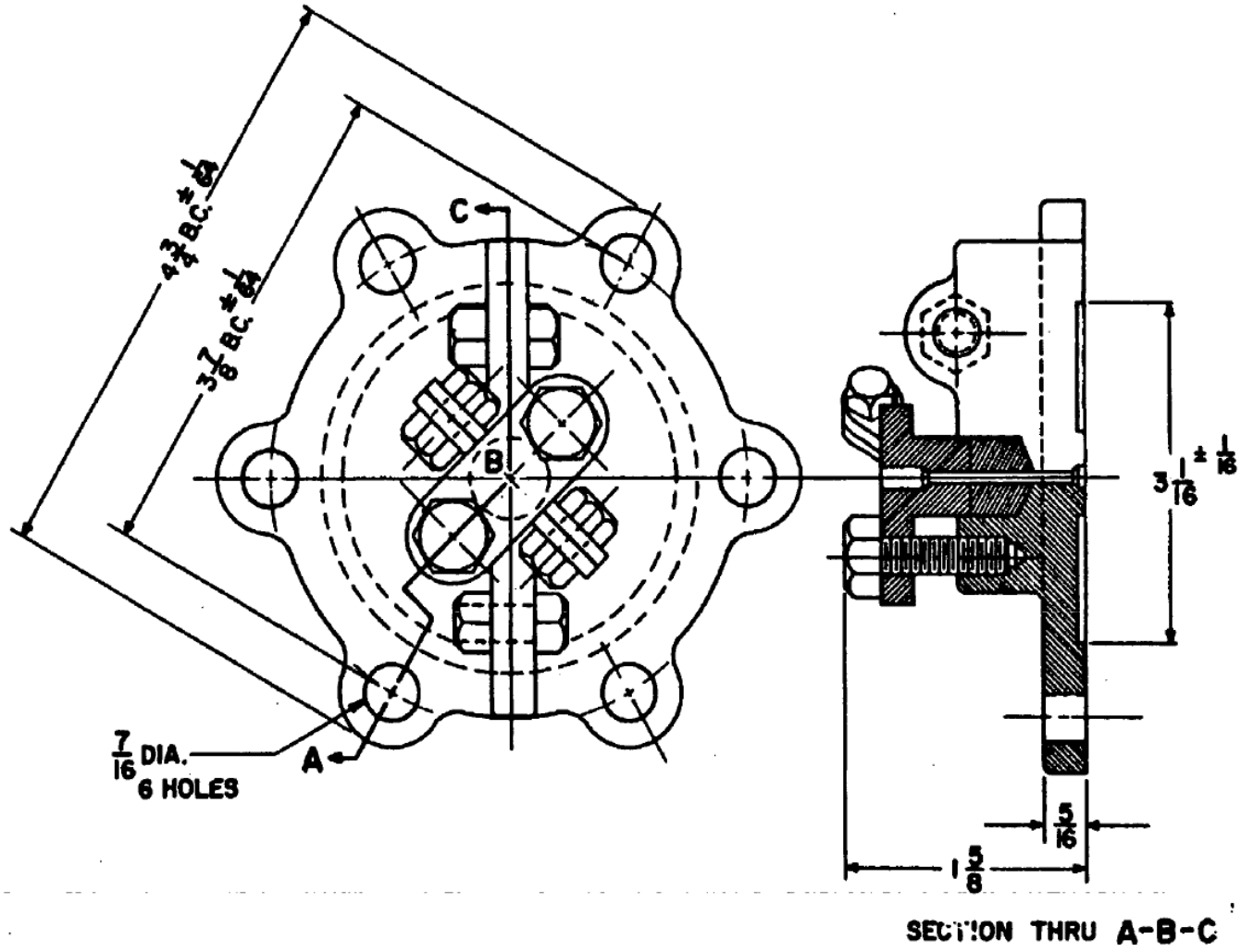


FIGURE 3. Bulkhead stuffing box.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:13Z
Check the source to verify that this is the current version before use.

3.7 Onboard repair parts. When specified (see 6.2), onboard repair parts shall be furnished in accordance with table V.

TABLE V. Onboard repair parts.

Repair parts	Quantity					
	Total number of identical parts installed per ship					
	1-10	11-20	21-30	31-40	41-50	Over 50
Thermostats	1	2	3	4	5	6
Seats and poppets	1	3	5	7	9	11
Manual shut-off elements and operating wrenches, if required	1	1	2	2	3	3

3.8 Identification and marking.

3.8.1 Identification plate. Each valve shall be provided with a metal identification plate, attached to an exposed position that normally will not be covered by insulation. The identification plate shall contain the following information:

a. Item name (thermostat type)
b. Valve size and number
c. Federal stock number
d. Component identification plate
e. Manufacturer's name
f. Manufacturer's address

3.8.2 Body marking. The valve size, manufacturer's name, and the direction of flow indicated by an arrow or similar means shall be cast or otherwise permanently marked on the valve body.

3.9 Workmanship. Temperature-regulating valves shall be free from defects that affect their appearance or their operation. Castings shall be clean, sound, and free from blow holes, hard spots, porosity, cracks, and other injurious defects. They shall be smooth and well-cleaned, both inside and outside, and all fins and roughness shall be removed.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall be performed on each model temperature-regulating valve at a laboratory acceptable to the Naval Sea Systems Command (NAVSEA). This inspection shall include the examination of 4.4 and tests of 4.5, as specified in table VI.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the tests of 4.5.3 through 4.5.5, as specified in table VI.

4.3.1 Sampling for conformance inspection.

4.3.1.1 Lot. All temperature-regulating valves of the same size and model and with the same type of thermostat(s) offered for delivery at one time shall be considered a lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:13Z
Check the source to verify that this is the current version before use.

4.3.1.2 Sampling for examination. A random sample of valves shall be selected from each lot for the examination specified in 4.4 with conformance based on the sampling requirements of ANSI/ASQ Z1.4, at inspection level III.

TABLE VI. Qualification and conformance inspections.

Characteristic	Requirement	Qualification	Conformance
Temperature-regulating valve	3.2.1	4.5	4.5.3, 4.5.4, 4.5.5
Thermostat	3.2.2	4.5.4	4.5.4
Materials	3.3	4.4	4.4
Removable parts	3.4.1	4.4	4.4
Shock	3.4.2	4.5.1	
Dimensions and tolerances	3.4.3	4.4	4.4
Pressure capacity	3.4.4	4.5.2.1, 4.5.3	4.5.3
Valve operation	3.4.5	4.5.2	
Valve unit	3.4.6	4.4	4.4
Thermostat	3.4.7	4.5.4	4.5.4
Bellows	3.4.8	4.4	4.4
Capillary tubing	3.4.9	4.4	4.4
Bulbs	3.4.10	4.4	4.4
Manually operable element	3.5	4.4	4.4
Bulkhead stuffing box	3.6	4.4	4.4
Workmanship	3.9	4.4	4.4

4.4 Examination. Each of the temperature-regulating valves tested in accordance with 4.5 or sampled for 4.3.1.2 shall be examined and measured to verify compliance with the requirements of this specification. Materials used in the manufacture of valves shall be examined to determine conformance to 3.3. This examination shall encompass all visual examinations and dimensional requirements specified in table VI. Classifications of defects are shown in table VII. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening the maximum efficiency shall constitute cause for rejection.

4.5 Test methods. Test methods shall be as listed in table VI. The requirements listed in table VI shall be tested in accordance with the applicable tests and test paragraphs listed therein. Results of tests shall be in accordance with the requirements paragraph listed in table VI or in the test paragraph.

4.5.1 High-impact shock test. Each type of thermostat and the largest size of each model temperature-regulating valve shall be shock tested in accordance with MIL-S-901 for Grade A, Type A, Class I equipment. The valve shall be tested in the open position and in the closed position. The test shall be conducted prior to the tests of 4.5.2 through 4.5.5; the correction of damage which may have occurred during shock tests shall not be performed prior to these tests. Cause for rejection shall be evidence of fragmentation or missile effect of parts, or deformation of parts that will cause active interference between parts.

4.5.2 Performance tests. Performance tests shall be conducted on one temperature-regulating valve of each model, size, and type of thermostat(s) as specified in 4.5.2.1 through 4.5.2.4, inclusive, and shall be made with dry saturated steam at a pressure of not less than 34 psig nor more than 36 psig at the valve inlet.

4.5.2.1 Valve capacity. Valve capacity shall be determined with a 25-psig back pressure at the valve outlet and with the valve fully open.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:13Z
Check the source to verify that this is the current version before use.

4.5.2.2 Valve opening point temperature. Valve opening point temperature shall be determined with the thermostats set at any setting within the adjustable range; the temperature at the thermostats shall then be lowered to 1 °F below the opening point to determine that there is a definite increase in the flow of steam.

4.5.2.3 Valve closing point temperature. Valve closing point temperature shall be determined with the thermostats set at any setting within the adjustable range; the temperature at the thermostats shall then be increased 1 °F above the closing point to determine that the valve will close fully and is steamtight.

4.5.2.4 Rate of condensate measurement. With a 25-psig back pressure at the valve outlet, the rate of condensate shall be measured when the temperature at the thermostat is 1 °F, 2 °F, and 3 °F below the opening point of the control valve.

4.5.2.5 Design change check test. When design changes are approved by NAVSEA for a particular size and model of valve or type of thermostat which may alter the performance of the temperature-regulating valve, tests specified in 4.5.2 shall be conducted on that size and model valve on the first item produced under the contract or order.

4.5.3 Pressure test (valves). Each valve body shall be tested with 150-psig air pressure while immersed in water or 150-psig hydrostatic pressure with the valve open to test the strength of the body and detect porosity. Visible deformation or porosity or other signs of weakness shall be cause for rejection.

4.5.4 Thermostat control point test. Each Type L and each Type R thermostat shall be tested to determine whether the fully open and fully closed points are within specified limits when thermostats are set at the mid-position of their adjustable temperature range. Each Type W thermostat shall be tested to determine whether the fully open point is as specified in 3.4.7.3.

4.5.5 Leakage test. The valve seats and poppets of each temperature-regulating valve shall be tested to determine the tightness and resistance to leakage at a minimum steam pressure of 50 psig on the inlet side of the valve and the outlet side open to the atmosphere. The test for valves with Type L or Type R thermostat shall be made with the thermostat set at the mid-position of the adjustable range in an ambient temperature from 3 to 6 °F above the set point.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:13Z
Check the source to verify that this is the current version before use.

TABLE VII. Classification of possible defects in accordance with ANSI/ASQ Z1.4.

Categories	Defects
Critical:	
1	None defined.
Major:	
101	Model and size of valve not as specified, or not packless type.
102	Thermostat damaged or not the specified model.
103	Materials defective or not as specified.
104	Operating bellows damaged or not protected as specified.
105	Failure to provide for easy removal of thermostat and replacement by manually operable element.
106	Union connections (inlet and outlet) damaged or not as specified.
107	Thermostat capillary tubing kinked or not protected by armor.
108	Thermostat mounting assembly damaged or not as specified.
109	Thermostat dial adjustment damaged or not provided with a means of locking at any setting in the operating range (Types L and R thermostats only).
110	Valve incomplete, parts missing or damaged.
Minor:	
201	Marking, manufacturer's name or trademark missing, not permanent, illegible or not as specified.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Thermostatic-controlled steam valves are for Navy shipboard ventilation heaters where service steam is available. These valves operate by self-regulating control, requiring no separate power source such as electricity or compressed air.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:13Z
Check the source to verify that this is the current version before use.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Model of valves and types of thermostats required (see 1.2).

c. When qualification is required (see 3.1).

d. Valve model and size and type of thermostat required (see 3.4.3, 3.4.4, and 3.4.5).

e. Manually controlled element required (see 3.5).

f. When bulkhead stuffing box is required (see 3.6).

g. Onboard repair parts required (see 3.7).

h. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 3155 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.3.1 Provisions governing qualification. Copies of SD-6, "Provisions Governing Qualification," may be obtained using the ASSIST Online database at https://assist.dla.mil.

6.4 Subject term (key word) listing.

Copper

Self-operated

Thermostat

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Navy – SH

Review activity:
DLA – CC

Preparing activity:
Navy – SH
(Project 4820-2013-005)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T17:13Z
Check the source to verify that this is the current version before use.